

2023



Accelerating the Ingestion of 33 TB/Day by 25% Using AWS Graviton Processor with Informatica

Learn how Informatica enhanced observability in cloud data management using AWS Graviton3 processors.

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Up to 25% increase
in ingestion rate

Up to 25% reduction
in search latency

Up to 20% reduction
in cost

33 TB
of daily data ingestion accelerated



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Overview

Artificial intelligence (AI)–powered enterprise cloud data management platform company [Info](#) improve the performance of its observability services and remain cost efficient. The business ingests about 33 TB of



data each day, and it must process all this data to verify that its applications are performing well. “Our observability system plays a central role for our product developers, operations teams, and customers,” says Amreth Chandrasehar, director of observability, site reliability engineering, machine learning (ML), and AI at Informatica. “We obviously seek high performance and cost optimization for this core solution.”

Starting in 2017, Informatica built and hosted many of its systems on Amazon Web Services (AWS) to optimize performance for its internal teams and customers. In 2021, Informatica built its observability system on [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service to run Kubernetes. The solution was built on top of [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity.



Initially, Informatica implemented x86-based Amazon EC2 worker nodes to unlock scale without driving up costs across its observability service. Then, when AWS released the third generation of its [AWS Graviton processor](#)—which is designed to deliver the optimal price performance for cloud workloads running on Amazon EC2—Informatica migrated its observability system to AWS Graviton3-based instances. “Every time AWS releases a new instance type, we are probably the first ones to implement it,” says Chandrasehar. “We do an evaluation and, if it works for our needs, we make a quick migration as soon as possible.”

Thus, within 45 days of the release of AWS Graviton3 processors, Informatica migrated its entire observability infrastructure to AWS Graviton3-based instances, improving performance, optimizing costs, and increasing scalability.

Opportunity | Using AWS Graviton3–Based Instances to Reduce Latency by 25% for Informatica

Founded in 1993, Informatica helps its customers make the most of their data. Its platform and solutions are powered by AI to give users insight into complex operations. As the company consistently looks for ways to enhance performance, it aims for efficiency. The business handles AI-powered data at a massive scale, making close to 1.6 trillion documents available to internal users to facilitate data searching. Moreover, it requires 2.2 PB of storage to provide the data services that its customers need.

Before the migration to Amazon EC2 in 2021, the company took about 3–4 minutes to search 1 PB of data. Informatica substantially reduced that time by using AWS services. By early 2023, Informatica took about 20 seconds to search 2.2 PB of data. Using AWS Graviton3-based instances, the company reduced search times to 15 seconds. This is over 30 times faster than before the initial migration to Amazon EC2 in 2021. “Due to the performance improvements that we’ve achieved using AWS Graviton3-based instances, we can add more complex queries, implement checks faster, and catch system issues more efficiently, improving service levels for our customers,” says Chandrasehar.

Solution | Reducing Costs by up to 20% Using AWS Graviton Processor

Immediately after AWS released AWS Graviton3 processors in May 2022, Informatica started evaluating AWS Graviton3-based instances. It then swiftly migrated its observability infrastructure—composed of 750 instances—to the service. The project took a three-person team six weeks to complete, from proof of concept to delivery. Not only did Informatica reduce cost per instance, but the greater processing power of AWS Graviton3 processors also reduced the total instance count by about 10 percent. By consolidating its footprint, the company made its system more sustainable while reducing its observability spending by up to 20 percent.

Informatica runs its observability workload on top of Amazon EKS using a self-managed search solution from Elastic. It also uses [Amazon Managed Streaming for Apache Kafka](#) (Amazon MSK), a fully managed data streaming service, to collect data logs from various systems and ingest them into its search solution. In tandem, these services significantly increased the processing power of Informatica's critical observability system. "Using AWS Graviton3-based instances, we increased our ingestion rate by up to 25 percent and reduced search latency by up to 25 percent," says Chandrasehar. Greater processing power translates to better performance for both the company and its customers.

Performance enhancements have also opened the door to further innovation. "Because the data is now available much faster, we can apply AI/ML features on top of the data we collect," says Chandrasehar. In fact, the company has taken much of the cost savings on processing to invest in AI/ML-based applications. For example, Informatica uses [Amazon SageMaker](#), a service to build, train, and deploy ML models, to develop generative AI solutions. The goal is to continue evolving its services to deliver new levels of insight to customers.

Informatica is now scaling to meet marketplace demand using a cost-optimal approach. As its data platform continues to grow, the company is using AWS services to power both internal applications and the services it delivers to customers. "The reliability, availability, and performance of the AWS services that we are using today have made a substantial impact on our teams and customers," says Chandrasehar.

Outcome | Using AWS Services to Drive Insights and Deliver Better Solutions



Informatica is on a mission of continuous cloud and AI innovation, and migrating its observability system to AWS Graviton3-based instances is only one step on its journey. “We are always looking to take advantage of the new innovations released by AWS,” says Chandrasehar.

Today, Informatica is using its enhanced observability solution to deliver a higher level of service to customers. “We are using AWS innovations to apply AI/ML and deliver AI-powered data insights to our cloud customers,” says Chandrasehar.

About Informatica

Founded in 1993, Informatica provides a data management solution powered by artificial intelligence to more than 5,000 customers around the world. The company’s systems process more than 54 trillion cloud transactions each month.

AWS Services Used

AWS Graviton processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run containerized applications on AWS cloud and on-premises data centers.



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Amazon MSK

Amazon Managed Streaming for Apache Kafka (Amazon MSK) makes it easy to ingest and process streaming data in real time with fully managed Apache Kafka.

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